



MA385 Part 2: Initial Value Problems

2.2: Euler's Method

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For more details, see Chapter 6 of [Süli and Mayers, *An Introduction to Numerical Analysis*](#).

1. The goal

Our goal is to generate numerical solutions to initial value differential equations. A solution to such problems is a function, usually of one variable denoted t . Our approximation will give estimates of the values of this function at certain points.

We'll denote the points we at which we are seeking approximations as

$$t_0 < t_1 < \cdots < t_N.$$

The methods we'll use are all **one-step** methods, and the first example we'll consider is **Euler's Method**.

Although it is not too important, we'll make the assumption that the points are equally spaced. So, we choose N and a final time, t_N , and the set $h = (t_N - t_0)/N$. Then the points are

$$t_0, \quad t_1 = t_0 + h, \quad t_2 = t_0 + 2h, \dots, \quad t_N = t_0 + Nh.$$

2. Euler's Method: motivation

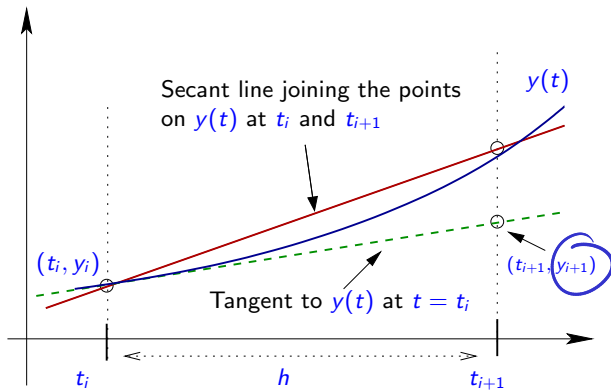
The simplest method is **Euler's Method**. We motivate it as follows.

Motivation

Suppose we know $y(t_i)$, and want to compute $y(t_{i+1})$. From the differential equation we can calculate the slope of the tangent to y at t_i . If this approximates the slope of the line joining $(t_i, y(t_i))$ and $(t_{i+1}, y(t_{i+1}))$, then

$$y'(t_i) = f(t_i, y(t_i)) \approx \frac{y_{i+1} - y_i}{t_{i+1} - t_i}.$$

2. Euler's Method: motivation



3. Euler's Method: formula

Euler's Method *(Important!)*

Choose equally spaced points $t_0, t_1, \dots, t_N$ so that

$$t_i - t_{i-1} = h = (t_N - t_0)/N \quad \text{for } i = 1, \dots, N.$$

We call h the “time step”. Let y_i denote the approximation for $y(t)$ at $t = t_i$. Set

$$y_{i+1} = y_i + hf(t_i, y_i), \quad i = 0, 1, \dots, N-1. \quad (1)$$

Motivation

$$\frac{y(t_{i+1}) - y(t_i)}{\underbrace{t_{i+1} - t_i}_h} \approx \frac{y'(t_i)}{f(t_i, y_i)} \Rightarrow y(t_{i+1}) \approx y(t_i) + hf(t_i, y_i)$$

Example 2.2.1

Taking $h = 1$, estimate $y(4)$ where

$$y'(t) = y/(1+t^2), \quad y(0) = 1.$$

$t_0 = 0$

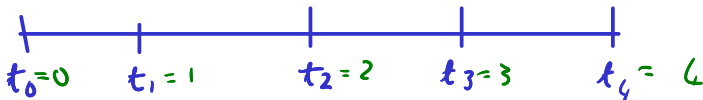
$y_1 = 1$

Choosing $h = 1$ we get

▶ $i = 0$: $t_0 = 0, y_0 = 1.$

▶ $i = 1$: $t_1 = t_0 + h = 1.$

$$y_1 = y_0 + hf(t_0, y_0) = 1 + \frac{1}{1+0^2} = 2.$$



▶ $i = 2: t_2 = t_0 + 2h = 2.$

$$y_2 = y_1 + hf(t_1, y_1) = 2 + 1 \frac{2}{1+1^2} = 3.$$

▶ $i = 3: t_3 = t_0 + 3h = 3.$

$$y_3 = y_2 + hf(t_2, y_2) = 3 + 1 \frac{3}{1+2^2} = 3.6$$

▶ $i = 4: t_N = t_4 = t_0 + 4h = 4.$

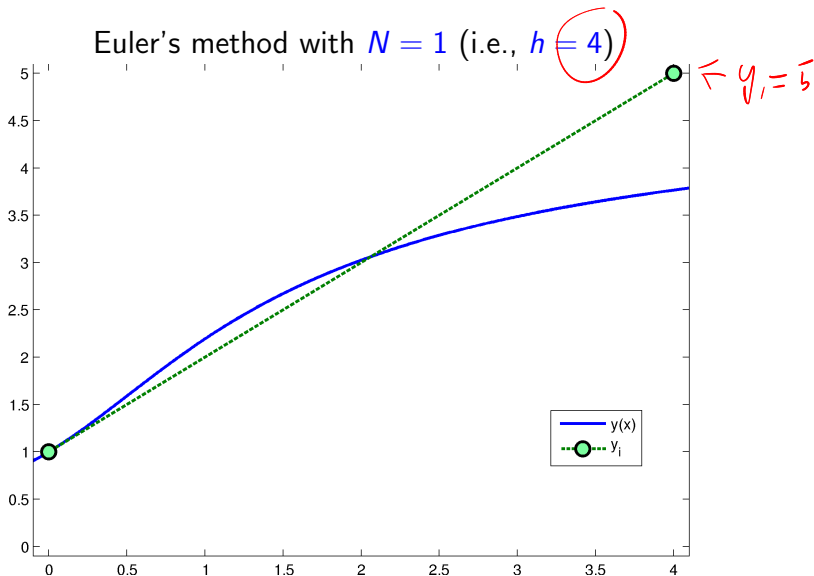
$$y_N = y_4 = y_3 + hf(t_3, y_3) = 3.6 + \frac{3.6}{1+3^2} = 3.96$$

Finished here Monday, 5 Oct (W5.1)

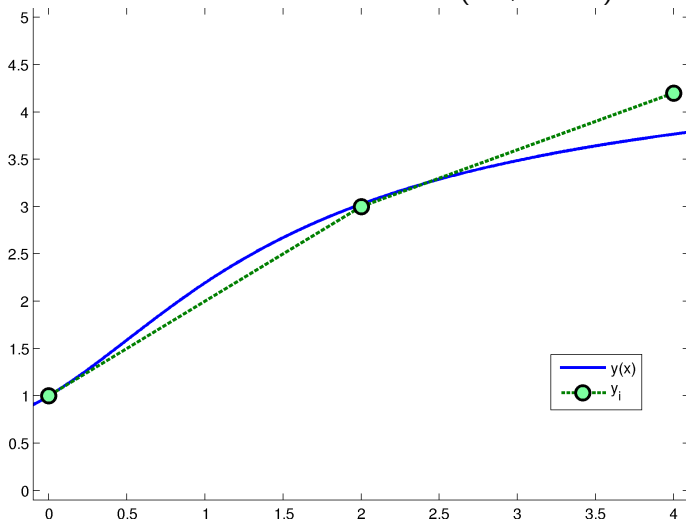
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If we had chosen $h = 4$ we would have only required one step:
 $y_N = y_0 + 4f(t_0, y_0) = 5$. However, this would not be very accurate.

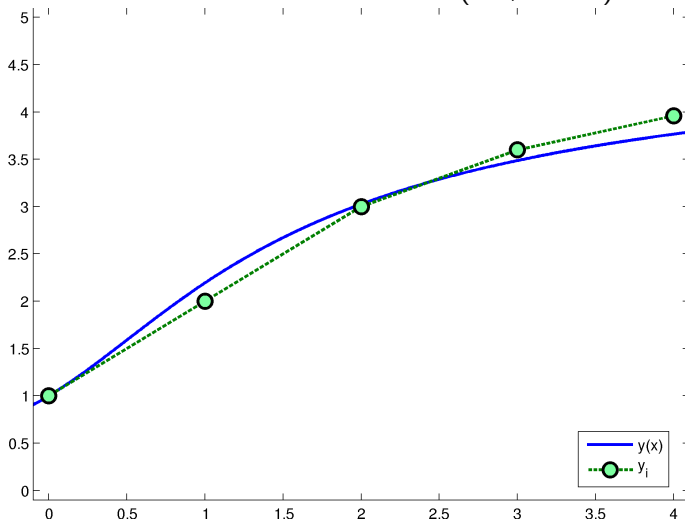
With a little work one can show that the solution to this problem is $y(t) = e^{\tan^{-1}(t)}$ and so $y(4) = 3.7652$. Hence the computed solution with $h = 1$ is much more accurate than the computed solution when $h = 4$. This is also demonstrated in next figure below, and in the follow table, where we see that the error seems to be proportional to h .

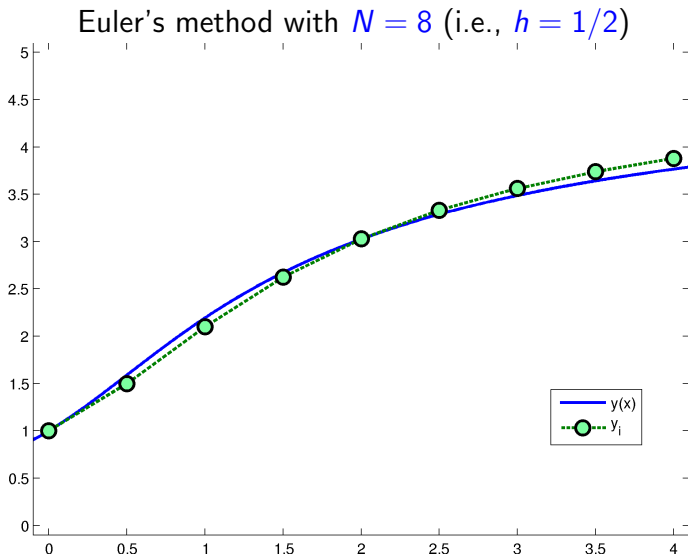


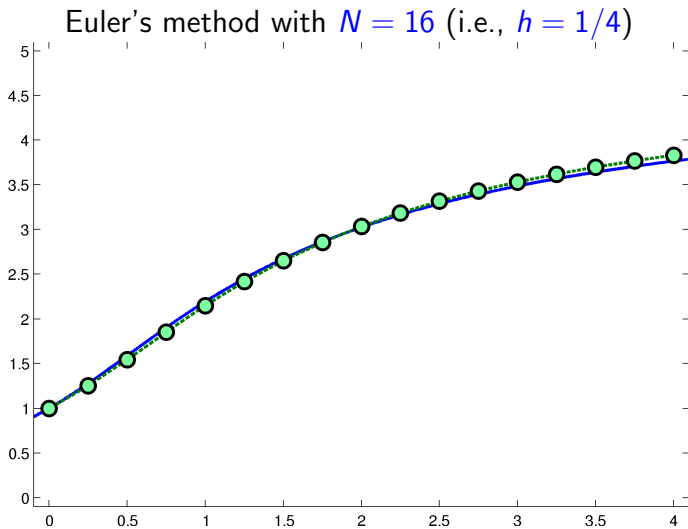
Euler's method with $N = 2$ (i.e., $h = 2$)

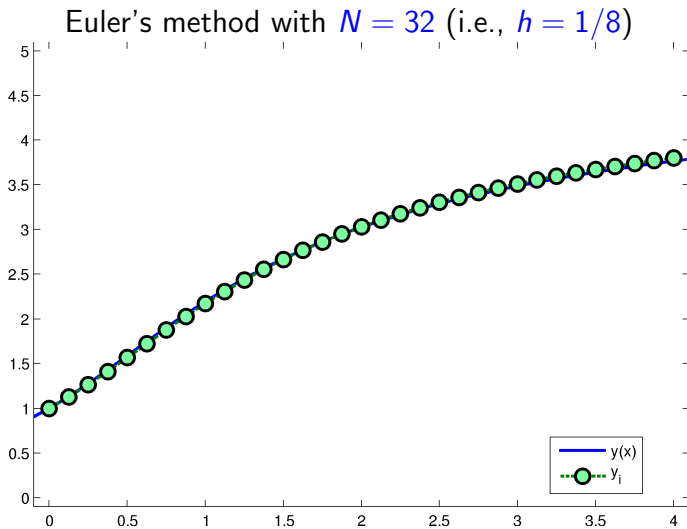


Euler's method with $N = 4$ (i.e., $h = 1$)









N	h	y_N	$ y(t_N) - y_N $
1	4	5.0	1.235
2	2	4.2	0.435
4	1	3.960	0.195
8	1/2	3.881	0.115
16	1/4	3.831	0.065
32	1/8	3.800	0.035

Table 1: Error in Euler's method for Example 7

4. Exercises

Exercise 2.2.1

As a special case in which the error of Euler's method can be analysed directly, consider Euler's method applied to

$$y'(t) = y(t), \quad y(0) = 1.$$

The true solution is $y(t) = e^t$.

- (i) Show that the solution to Euler's method can be written as

$$y_i = (1 + h)^{t_i/h}, \quad i \geq 0.$$

- (ii) Show that $\lim_{h \rightarrow 0} (1 + h)^{1/h} = e$. (Note that this shows that, if we denote by $y_n(T)$ the approximation for $y(T)$ obtained using Euler's method with n intervals between t_0 and T , then

$$\lim_{n \rightarrow \infty} y_n(T) = e^T.$$

Hint: Let $w = (1 + h)^{1/h}$, so that $\log w = (1/h) \log(1 + h)$. Now use l'Hospital's rule to find $\lim_{h \rightarrow 0} w$.